

Problem 1: Assembly production

A company produces assemblies containing one frame, one shaft and one purchased ball bearing. Shafts require 0.6 hour of forging, 0.3 hour on a lathe and 0.4 hour of grinding. Frames require 0.8 hour of forging, 0.2 hour of drilling, 0.3 hour of milling and 0.6 hour of grinding. The company has 20 forging machines, 5 lathes, 10 grinders, 3 drillers and 6 millers. Each machine operates at most 4500 hours per year. Formulate an LP maximizing annual assembled output.

Decision variables

- x_1 : shafts produced per year.
- x_2 : frames produced per year.

Each assembly needs one shaft and one frame, so $x_1 = x_2$. Ball bearings are purchased.

All processing coefficients are hours per part. Annual capacity is the number of machines multiplied by 4500 hours.

Machine	Hours/shaft	Hours/frame	Machines	Annual hours
Forging	0.6	0.8	20	90,000
Drilling	0	0.2	3	13,500
Milling	0	0.3	6	27,000
Lathe	0.3	0	5	22,500
Grinding	0.4	0.6	10	45,000

Because shaft and frame production are equal, x_1 is also the number of complete assemblies.

$$\begin{aligned}
 \max \quad & Z = x_1 \\
 \text{subject to} \quad & x_1 - x_2 = 0, \\
 & 0.6x_1 + 0.8x_2 \leq 90000, \\
 & 0.2x_2 \leq 13500, \\
 & 0.3x_2 \leq 27000, \\
 & 0.3x_1 \leq 22500, \\
 & 0.4x_1 + 0.6x_2 \leq 45000, \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

Problem 2: Chicken-feed blending

An agricultural mill mixes corn, limestone and alfalfa to produce exactly N bushels of chicken feed. Their costs are respectively 12, 5 and 8 dollars per bushel. Ingredient i supplies c_{ij} units of nutrient j per bushel, where $i = 1, 2, 3$ denotes corn, limestone and alfalfa, and $j = 1, 2, 3, 4$ denotes protein, calcium, carbohydrates and vitamins. Each bushel of finished feed must contain at least P_j and at most u_j units of nutrient j . Formulate an LP minimizing cost while meeting the quantity and nutritional requirements.

Decision variables

- x_1 : bushels of corn purchased and blended, at 12 dollars per bushel.
- x_2 : bushels of limestone purchased and blended, at 5 dollars per bushel.
- x_3 : bushels of alfalfa purchased and blended, at 8 dollars per bushel.

N is the required batch size. c_{ij} is the nutrient content per bushel of ingredient i ; P_j and u_j are the lower and upper nutrient limits per bushel of feed.

$$\begin{aligned} \min \quad & Z = 12x_1 + 5x_2 + 8x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_3 = N, \\ & c_{11}x_1 + c_{21}x_2 + c_{31}x_3 \geq P_1N, \\ & c_{11}x_1 + c_{21}x_2 + c_{31}x_3 \leq u_1N, \\ & c_{12}x_1 + c_{22}x_2 + c_{32}x_3 \geq P_2N, \\ & c_{12}x_1 + c_{22}x_2 + c_{32}x_3 \leq u_2N, \\ & c_{13}x_1 + c_{23}x_2 + c_{33}x_3 \geq P_3N, \\ & c_{13}x_1 + c_{23}x_2 + c_{33}x_3 \leq u_3N, \\ & c_{14}x_1 + c_{24}x_2 + c_{34}x_3 \geq P_4N, \\ & c_{14}x_1 + c_{24}x_2 + c_{34}x_3 \leq u_4N, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

The nutrient limits are per bushel; multiplying by N gives the limits for the entire batch.

Problem 5: Polly production and inventory

Plan perfume production for January through April. Demand is 4000, 5000, 6000 and 4500 gallons, respectively. Initial inventory is 2000 gallons. Production costs 5 dollars per gallon in January and February and 5.50 dollars per gallon in March and April. Carrying inventory from one month to the next costs 1 dollar per gallon. All Polly must be sold by the end of April because a new perfume is introduced in May. Production in a month may meet demand in that same month. Formulate an LP minimizing production and inventory costs while meeting all demand.

Data and decision variables

D_t is demand in month t (gallons); C_{It} is the production cost in that month (dollars per gallon).

Month t	Demand D_t (gal)	Production cost C_{It} (dollars/gal)
1: January	$D_1 = 4000$	$C_{I1} = 5.00$
2: February	$D_2 = 5000$	$C_{I2} = 5.00$
3: March	$D_3 = 6000$	$C_{I3} = 5.50$
4: April	$D_4 = 4500$	$C_{I4} = 5.50$

Holding cost is $h = 1$ dollar per gallon carried to the next month. Initial stock is $I_0 = 2000$ gallons; required final stock is $I_4 = 0$.

- x_1, x_2, x_3, x_4 : gallons produced in January, February, March and April, respectively.
- I_1, I_2, I_3, I_4 : gallons held at the end of January, February, March and April, respectively.

Production and inventory are nonnegative decision variables. D_t , C_{It} , h and I_0 are fixed data.

$$\begin{aligned}
 \min \quad & Z = 5(x_1 + x_2) + 5.5(x_3 + x_4) + 1(I_1 + I_2 + I_3) \\
 \text{subject to} \quad & 2000 + x_1 - I_1 = 4000, \\
 & I_1 + x_2 - I_2 = 5000, \\
 & I_2 + x_3 - I_3 = 6000, \\
 & I_3 + x_4 - I_4 = 4500, \\
 & I_4 = 0, \\
 & x_1, x_2, x_3, x_4 \geq 0, \quad I_1, I_2, I_3, I_4 \geq 0.
 \end{aligned}$$

Problem 6: Operations Research and optimization

Hello, Mr. Blank.

In life, we are always trying to achieve the best possible result. We want to produce more, spend less, work faster, and make better decisions. If there were no limitations, doing this would be easy. But in the real world, there are always constraints: limited money, limited time, limited land, limited energy, and limited resources.

With just those two ideas, having an objective and having limitations, we already have the foundation of Operations Research and optimization. The basic question is simple: given the resources that we have, how can we use them in the best possible way to achieve our goal?

Let me give you an example related to energy.

Imagine that we need to supply enough renewable energy to a small city using solar panels and wind turbines. We know how much energy the city needs, so we have a minimum energy requirement that must be satisfied.

However, we also face several limitations. Wind turbines may produce more energy, but they can be more expensive and require more land. Solar panels may be cheaper or easier to install, but we may need many more of them to produce the same amount of energy. At the same time, we have a limited budget and only a certain amount of land available.

So the question becomes: how many solar panels and wind turbines should we install? How should we allocate our available land and money so that the city receives the energy it needs while using those resources as efficiently as possible?

That is an optimization problem.

And this same idea appears again and again in the Department of Energy: deciding how to allocate resources, how to operate systems, how to schedule production, how to design energy networks, and how to make better decisions when several constraints must be satisfied at the same time.

Optimization gives us mathematical tools, algorithms, and software that help analyze these decisions systematically instead of relying only on intuition.

In life, we are always trying to achieve the best possible outcome. The challenge is that we never have unlimited resources. Optimization is the science of finding the best possible decision within those limits.